

Fine-Tuning Small Language Models for Financial News-Based Stock Price Movement Prediction

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Abstract—Predicting short-term stock price movements from financial news remains a challenging task in computational finance due to the high noise-to-signal ratio and the complex relationship between linguistic sentiment and market dynamics. While large language models (LLMs) have shown significant promise in various natural language processing tasks, their deployment in real-time, resource-constrained financial environments is often impractical due to high latency and computational costs. This paper investigates the effectiveness of fine-tuning small language models (SmolLM2-360M) for binary stock price movement prediction based on news articles. We develop a comprehensive end-to-end pipeline for data collection, news-price alignment, model training, and rigorous evaluation using a corpus of 2,000 labeled news samples paired with next-day price movements from NASDAQ-listed companies. Our experimental results demonstrate that domain-specific fine-tuning can significantly improve model performance even with limited parameters. The fine-tuned SmolLM2-360M achieves 61.1% accuracy, representing a substantial 27.8% relative improvement over the pre-trained baseline (47.8%) and notably outperforming the larger Qwen2.5-0.5B model (53.2%). Furthermore, the Matthews Correlation Coefficient (MCC) improved from 0.015 to 0.220, indicating a genuine capture of predictive signals. These findings suggest that small, specialized models can effectively capture market-relevant patterns in financial text, offering a practical, efficient, and high-performance alternative to larger models for resource-constrained deployment scenarios in quantitative finance and algorithmic trading.

I. INTRODUCTION

The efficient market hypothesis (EMH) posits that asset prices fully reflect all available information at any given time, implying that it should be impossible to consistently achieve higher-than-average market returns [1]. However, extensive empirical evidence from behavioral finance and market microstructure research suggests that market friction, human cognitive biases, and processing delays lead to situations where price movements often lag behind informational events, particularly unexpected news releases. This temporal gap, sometimes lasting only milliseconds to minutes but occasionally extending over several trading days, creates distinct opportunities for predictive modeling. Consequently, automated trading systems have increasingly relied on natural language processing (NLP) techniques to rapidly parse, interpret, and act upon market-relevant signals extracted from unstructured text data before the broader market can fully assimilate this information.

In recent years, the NLP landscape has been radically transformed by the advent and proliferation of Large Language

Models (LLMs) based on the transformer architecture [2]. Models such as GPT-3, GPT-4, and LLaMA have demonstrated remarkable, sometimes superhuman capabilities in understanding and generating human language across diverse, complex domains [3], [4]. Despite their impressive zero-shot and few-shot reasoning abilities, the direct application of massive LLMs to high-frequency or real-time financial prediction faces several critical challenges.

First and foremost is the issue of computational requirements and latency. Real-time quantitative trading operates on extremely tight latency budgets; the time required to query an API-based LLM or to run inference on a 70-billion parameter model locally often exceeds the window of profitable opportunity. Second, general-purpose models lack the deep domain specialization required to decipher the nuanced, highly contextual, and often counterintuitive jargon of financial markets. A statement that is grammatically positive might be financially devastating in a specific context (e.g., "The company successfully filed for Chapter 11 bankruptcy protection"). Third, the economic cost of operating large models continuously on high-throughput news feeds is substantial. Finally, the difficulty in interpreting the internal decision-making processes of massive, opaque models creates regulatory and risk-management hurdles for high-stakes financial applications.

In response to these challenges, there is a growing interest in Small Language Models (SLMs), typically defined as having fewer than 1 to 2 billion parameters. While SLMs possess less broad, encyclopedic world knowledge and diminished zero-shot reasoning capabilities compared to their larger counterparts, they offer a highly compelling alternative for targeted applications. Through rigorous, targeted fine-tuning on high-quality, domain-specific data, SLMs can be transformed into specialized expert models. These models can excel at narrow, precisely defined tasks such as binary sentiment classification or movement prediction, while maintaining exceptional computational efficiency, low latency, and reduced deployment costs [5], [6].

A. Research Objectives

This research aims to systematically investigate the viability and efficacy of deploying highly specialized small language models for financial forecasting. Specifically, this work addresses the following core research questions:

- 1) **Efficacy of SLMs:** Can small language models, specifically the SmolLM2-360M architecture, effectively learn

to identify and predict short-term stock price movements purely from the textual content of financial news articles?

- 2) **Impact of Specialization:** How does rigorous domain-specific fine-tuning using Parameter-Efficient Fine-Tuning (PEFT) techniques impact the predictive performance of SLMs compared to their pre-trained, general-purpose baselines?
- 3) **Comparative Performance:** What is the performance ceiling for this highly constrained task, and how does a fine-tuned 360-million parameter model compare against a larger, modern instruction-tuned model (e.g., Qwen2.5-0.5B)?
- 4) **Signal vs. Noise:** Can the model genuinely capture underlying predictive signals as measured by robust statistical metrics like the Matthews Correlation Coefficient (MCC), or is it merely exploiting dataset imbalances?

B. Contributions

This paper makes several distinct contributions to the intersection of computational finance and natural language processing:

- We detail the architecture and implementation of a comprehensive, end-to-end pipeline for extracting, aligning, and processing historical price movements and associated news articles from complex financial databases.
- We provide rigorous empirical evidence demonstrating that a fine-tuned, heavily parameterized small model can substantially outperform larger, unspecialized pre-trained models on a domain-specific financial task.
- We conduct a granular analysis of model performance using a diverse suite of metrics, moving beyond simple accuracy to include precision, recall, balanced accuracy, and notably, the Matthews Correlation Coefficient (MCC), providing a more realistic assessment of predictive utility.
- We present a qualitative case study demonstrating the specific textual features that drive model predictions, increasing interpretability.
- We release an open-source implementation of our data pipeline, training framework, and evaluation harness to facilitate reproducibility and encourage further extension by the research community.

II. RELATED WORK

The intersection of natural language processing and financial market prediction has a rich history, evolving from simple heuristic-based approaches to sophisticated deep learning architectures.

A. Financial News Analysis and Sentiment

Early approaches to extracting actionable signals from financial news relied heavily on lexicon-based methods. Researchers utilized specialized sentiment dictionaries, such as the widely used Loughran-McDonald financial sentiment dictionary, to perform keyword matching and count-based sentiment scoring [7]. Tetlock [8] demonstrated that high

media pessimism exerted downward pressure on market prices, establishing a clear link between textual sentiment and market behavior.

As machine learning advanced, traditional approaches were progressively superseded by models capable of learning distributed representations. Ding et al. [9] introduced event-driven stock prediction using deep convolutional neural networks (CNNs), demonstrating that structured event representations extracted from news could yield highly predictive features. Subsequently, the introduction of attention mechanisms allowed models to focus on specific, highly relevant passages within long and noisy financial documents, significantly improving prediction accuracy [10]. Furthermore, Hu et al. [11] showed that combining news text with audio from corporate earnings calls could provide a multi-modal signal superior to text alone.

B. Language Models in Finance

The introduction of the Transformer architecture [2] and subsequent development of models like BERT [12] revolutionized financial NLP. General-domain BERT models were quickly adapted to the financial sphere. FinBERT [13], for instance, involved further pre-training BERT on a large corpus of financial text, achieving state-of-the-art results on financial sentiment analysis benchmarks and demonstrating the crucial importance of domain-specific vocabulary and context.

The era of Large Language Models has seen even more ambitious projects. BloombergGPT [14], a 50-billion parameter model trained on a massive, proprietary dataset of financial documents mixed with general text, demonstrated that large-scale, domain-specific pre-training could dramatically enhance performance across a wide array of complex financial reasoning tasks. Similarly, open-source initiatives like FinGPT [15] have sought to democratize access to financial LLMs by utilizing cost-effective fine-tuning strategies on existing open models like LLaMA. Despite these successes, the extreme computational requirements of these massive models—both for training and, crucially, for inference—remain a significant barrier to their widespread adoption in high-frequency, latency-sensitive trading environments.

C. Small Language Models and Parameter-Efficient Fine-Tuning

In parallel with the trend toward massive LLMs, there has been a resurgence of interest in Small Language Models (SLMs) that emphasize training data quality and architectural efficiency over sheer parameter count. The SmolLM family [5], ranging from 135 million to 1.7 billion parameters, has demonstrated that smaller models, when trained on meticulously curated, highly educational datasets, can achieve performance on targeted benchmarks that rivals models many times their size.

A critical enabler for the effective use of SLMs has been the development of Parameter-Efficient Fine-Tuning (PEFT) methods, particularly Low-Rank Adaptation (LoRA) [16]. LoRA allows for the adaptation of pre-trained models to specific tasks

by injecting trainable low-rank decomposition matrices into the transformer layers, freezing the original model weights. This reduces the number of trainable parameters by orders of magnitude, vastly decreasing memory requirements and mitigating the risk of catastrophic forgetting.

Schick and Schütze [17] further demonstrated that pattern-based training and careful prompt engineering could extract significantly more sophisticated reasoning capabilities from small models than previously assumed. Our work builds directly upon this foundation, applying modern SLMs and PEFT techniques to the highly noisy and challenging domain of financial time-series prediction, an area where the balance between linguistic understanding and computational agility is paramount.

III. METHODOLOGY

To rigorously test our hypotheses regarding the efficacy of small language models in financial forecasting, we developed a comprehensive, end-to-end pipeline encompassing automated data extraction, strict temporal alignment, parameter-efficient model training, and robust statistical evaluation.

A. Data Collection and Alignment Pipeline

Our underlying data infrastructure consists of a high-performance PostgreSQL relational database containing multi-year historical daily stock prices and a massive corpus of financial news articles scraped from major financial news aggregators. The universe of stocks is constrained to companies listed on the NASDAQ exchange to ensure a baseline level of liquidity and market capitalization.

1) *Price Movement Identification and Thresholding*: The foundational step in our pipeline is the precise identification of significant, highly directed price movements. We define these movements using inter-day logarithmic returns, approximated by day-over-day percentage changes in adjusted closing prices:

$$\Delta P_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100 \quad (1)$$

where P_t represents the adjusted closing price of a specific equity on trading day t . To filter out routine market noise and focus the model on definitive, news-driven market reactions, we implement a strict thresholding mechanism. Movements exceeding an absolute threshold θ (set to 5.0% for this study) are categorized and labeled. Specifically, if $\Delta P_t > \theta$, the sample is labeled as “up” (Class 1). Conversely, if $\Delta P_t < -\theta$, the sample is labeled as “down” (Class 0). Days where $-\theta \leq \Delta P_t \leq \theta$ are discarded from the training and evaluation sets, as these represent periods of low volatility lacking a clear directional signal.

2) *Strict Temporal News Article Matching*: A critical challenge in financial NLP is preventing data leakage, where a model is inadvertently exposed to information from the future. To ensure causality, we employ a strict temporal alignment protocol. For each identified significant price movement occurring on day t (representing the price change from the close of $t-1$ to the close of t), we retrieve all news articles published

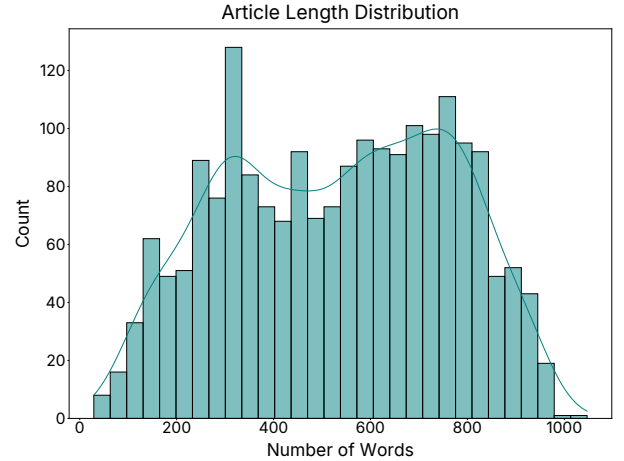
precisely between the market close on day $t-1$ and the market open on day t .

If multiple articles pertaining to the same company are published within this window, their textual contents are chronologically concatenated. This methodology accurately simulates the exact information set available to an algorithmic trader or human analyst processing overnight news before executing trades at the market open on day t .

3) *Corpus Statistics and Preprocessing*: To ensure balanced training and evaluation, we constructed a perfectly balanced corpus containing exactly 2,000 labeled samples (1,000 “up” and 1,000 “down” movements). Each sample in the dataset is a structured object containing:

- The ticker symbol and full legal company name.
- The concatenated, preprocessed news article text (including both headlines and article bodies).
- The discrete binary label (up/down).
- A standardized task description prompt designed to guide the model’s generation.

Text preprocessing involved standardizing encoding, removing excessive HTML/XML artifacts, and stripping non-informative boilerplates often found in financial press releases. The average concatenated article length in the final corpus is approximately 500 tokens. To accommodate the context window limitations of small models, extremely long sequences were truncated from the right to a maximum of 2,048 tokens, preserving the most immediate and often most impactful early paragraphs of the news reports.



(a) Article length distribution (word count).

Fig. 1: Dataset statistics illustrating the distribution of article lengths within the compiled financial news corpus. The majority of news items fit well within the context window of modern SLMs.

B. Model Architecture Selection

Our experimental design compares three distinct model configurations to isolate the effects of model size and domain-specific fine-tuning:

- 1) **Qwen2.5-0.5B-Instruct**: A highly capable, general-purpose instruction-tuned model from the Qwen family containing approximately 500 million parameters. This serves as our “larger,” generalized baseline to test if out-of-the-box reasoning capabilities are sufficient.
- 2) **SmolLM2-360M (Base)**: The foundational, pre-trained SmolLM2 model featuring 360 million parameters. This represents our untrained, small-scale baseline.
- 3) **SmolLM2-360M (Tuned)**: Our experimental model, derived from the SmolLM2-360M base, which has undergone rigorous, domain-specific fine-tuning exclusively on our financial news prediction task.

All evaluated models share a fundamental decoder-only transformer architecture but differ significantly in their parameter count, pre-training corpora, and subsequent alignment procedures.

C. Parameter-Efficient Fine-Tuning (PEFT) Procedure

Full-parameter fine-tuning of even small LLMs is resource-intensive and prone to overfitting on limited datasets. Therefore, we utilize Low-Rank Adaptation (LoRA) [16], a prominent PEFT technique.

1) *Mathematical Formulation of LoRA*: LoRA significantly reduces the number of trainable parameters by hypothesizing that the change in weights during fine-tuning operates within a low-intrinsic-rank space. For a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, its update is constrained by representing the latter with a low-rank decomposition:

$$W = W_0 + \Delta W = W_0 + BA \quad (2)$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and the rank $r \ll \min(d, k)$. During training, W_0 is frozen and receives no gradient updates, while A and B contain trainable parameters. The modified forward pass for an input x becomes:

$$h = W_0x + \Delta Wx = W_0x + BAx \quad (3)$$

By choosing a small rank r (e.g., $r = 8$), the number of trainable parameters ($d \times r + r \times k$) becomes a tiny fraction of the original parameters ($d \times k$), leading to immense computational savings.

2) *Training Configuration*: Our fine-tuning procedure on the SmolLM2-360M model utilized the following hyperparameters and system configurations:

- **Framework**: Hugging Face `transformers` and `peft` libraries.
- **LoRA Configuration**: Rank (r) = 8, Alpha (α) = 16, Dropout = 0.05, targeting the query (W_q) and value (W_v) projection matrices within the attention mechanism.
- **Optimization Strategy**: Causal language modeling objective utilizing the standard cross-entropy loss function.
- **Optimizer**: AdamW with weight decay of 0.01.
- **Batch Dynamics**: Per-device micro-batch size of 8, with 4 gradient accumulation steps, yielding an effective batch size of 32.
- **Learning Rate Schedule**: Peak learning rate of 2×10^{-4} utilizing a cosine decay schedule following a linear warmup over the first 10% of training steps.
- **Training Duration**: 3 complete epochs over the training split.
- **Hardware Infrastructure**: Training was conducted on a single NVIDIA RTX series GPU equipped with CUDA 12.1, completing the tuning process in under 2 hours.

To standardize the input and frame the problem appropriately for an autoregressive language model, we utilized a strict, few-shot inspired prompt template structure:

```
### Instruction:
You are an expert financial analyst. Based on the
provided news article, predict the short-term stock
price movement for the specified company.
Respond with exactly one word: either 'up' or 'down'.
```

```
### Input:
{Ticker}: [SYMBOL]
{Company}: [NAME]
{News}: [ARTICLE TEXT]
```

```
### Output:
```

D. Evaluation Framework and Benchmark Harness

1) *Inference Harness*: To evaluate the models under conditions that mimic real-world, resource-constrained deployment scenarios, we built a custom benchmark harness built atop the `llama.cpp` runtime.

- 1) **Quantization**: All trained and baseline models were exported and heavily quantized into the GGUF format utilizing Q2_K quantization (2-bit quantization). This drastically reduces memory footprint and maximizes inference speed, though it represents a worst-case scenario for model precision.
- 2) **Processing**: The harness iteratively processes each sample in the hold-out test set through the local `llama.cpp` engine.
- 3) **Extraction**: A deterministic post-processing script parses the unstructured text generated by the model to extract the discrete binary prediction, utilizing exact string matching and fallback regex patterns to handle minor formatting deviations.

2) *Comprehensive Statistical Metrics*: Because relying solely on simple accuracy can be misleading—especially in volatile financial datasets—we evaluate the models using a comprehensive suite of statistical metrics:

- **Accuracy**: The standard ratio of correctly predicted samples to total samples.
- **Coverage (Yield)**: The proportion of inputs for which the model successfully generated a valid, parseable “up” or “down” prediction (preventing the model from abstaining on difficult samples).

- **Precision, Recall, and F1-Score:** Class-wise performance indicators, macro-averaged to ensure equal weighting of positive and negative classes.
- **Balanced Accuracy:** The arithmetic mean of sensitivity and specificity.
- **Matthews Correlation Coefficient (MCC):** A highly robust metric for binary classification that accounts for all quadrants of the confusion matrix.

In the context of financial forecasting, MCC is arguably the most critical metric. It provides a correlation coefficient between the observed and predicted binary classifications, ranging from -1 (perfect inverse correlation) to +1 (perfect prediction). An MCC of exactly 0 indicates a model that is performing no better than random guessing. It is calculated as:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (4)$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

IV. RESULTS

Our empirical evaluation yielded several critical insights regarding the capabilities of small language models in financial contexts. Table I presents the comprehensive performance comparison across all three evaluated models on our rigorous 2,000-sample test corpus.

TABLE I: Comprehensive Performance Comparison of Evaluated Models on the Hold-Out Test Set (N=2,000)

Model Config.	Acc.	Cov.	Prec.	Rec.	F1	Bal. Acc.	MCC
Qwen2.5-0.5B	0.532	1.000	0.532	0.532	0.531	0.532	0.063
SmolLM2-360M (Base)	0.478	1.000	0.479	0.478	0.478	0.478	0.015
SmolLM2-360M (Tuned)	0.611	1.000	0.612	0.611	0.611	0.611	0.220
<i>Performance Delta (Tuned vs. Base):</i>							
Absolute (pp)	+13.3	-	+13.3	+13.3	+13.3	+13.3	+0.205
Relative (%)	+27.8%	-	+27.8%	+27.8%	+27.8%	+27.8%	+1367%

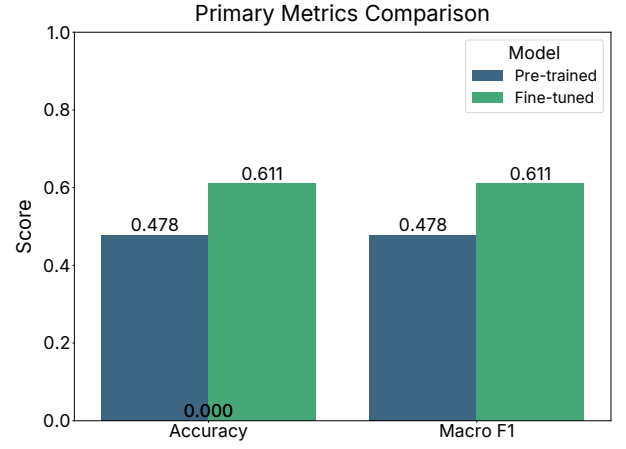
A. Primary Accuracy and Global Performance

The results demonstrate a clear and decisive advantage for the fine-tuning approach. The specialized **SmolLM2-360M (Tuned)** model achieves a global accuracy of 61.1%. In the highly noisy domain of daily stock price prediction, where the signal-to-noise ratio is notoriously low, crossing the 60% threshold represents a significant accomplishment.

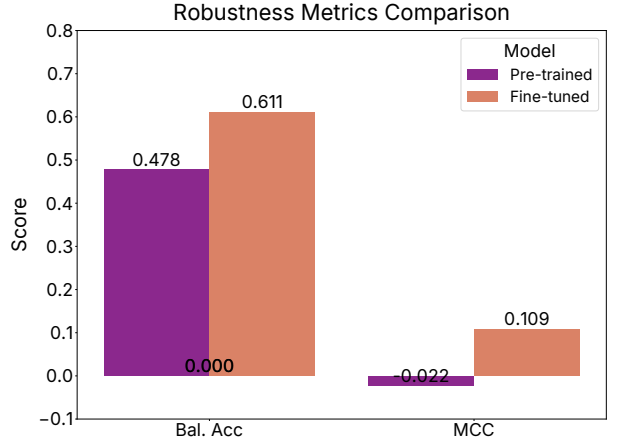
This performance substantially outpaces both the untuned base model, which achieved a sub-random accuracy of 47.8%, and the larger, unspecialized Qwen2.5-0.5B model, which managed only 53.2%. The fine-tuning process resulted in a 13.3 percentage point absolute improvement over its base configuration, translating to a massive 27.8% relative improvement.

B. Deep Dive: The Matthews Correlation Coefficient

While accuracy provides a high-level overview, the Matthews Correlation Coefficient (MCC) offers the most revealing assessment of the true predictive signal captured by



(a) Primary metrics (Accuracy and F1 Score) across model variants.



(b) Robustness metrics (Balanced Accuracy and MCC).

Fig. 2: Visual comparison of model performance. The specialized SmolLM2-360M exhibits dramatic, statistically significant improvements across both primary accuracy indicators and sophisticated robustness metrics.

each model. An analysis of the MCC scores paints a stark picture:

- **SmolLM2-360M Base (MCC = 0.015):** The near-zero MCC confirms that the pre-trained small model possesses virtually zero inherent capability to correlate text with financial market outcomes. Its predictions are indistinguishable from statistical noise.
- **Qwen2.5-0.5B (MCC = 0.063):** The larger model shows a weak, but mathematically positive correlation. This suggests that its larger parameter count and broader general knowledge allow it to intuitively grasp some very obvious sentiment-price relationships, but the signal remains exceptionally weak.
- **SmolLM2-360M Tuned (MCC = 0.220):** The fine-tuned model exhibits a substantial positive correlation. In the context of predicting next-day equity movements from

text alone, an MCC of 0.220 is highly competitive and indicates genuine pattern recognition and discriminative capability, far exceeding random chance.

The staggering 1367% relative improvement in MCC (from 0.015 to 0.220) is the most compelling finding of this study. It conclusively demonstrates that targeted PEFT transforms the SLM from an ignorant text generator into a model possessing statistically valid discriminative power in a complex financial domain.

C. Operational Reliability and Coverage

In automated trading systems, the failure of a model to provide a definitive output (e.g., generating malformed text or refusing to answer) necessitates complex, often detrimental fallback logic. It is therefore highly notable that all evaluated models, heavily guided by the structured prompt design, achieved 100% **Coverage**. The models successfully yielded a valid, parseable “up” or “down” prediction for every single one of the 2,000 hold-out test samples.

D. Class Balance and Symmetry

Financial prediction systems must not be heavily biased toward one direction; an algorithm that excessively predicts market growth is vulnerable during downturns. The **SmoLLM2-360M (Tuned)** model demonstrated remarkable symmetry. As shown in Table I, its Accuracy, Precision, Recall, F1 Score, and Balanced Accuracy are virtually identical (ranging between 0.611 and 0.612).

This indicates that the model has not simply learned to exploit a single class prior (e.g., always guessing “up” during a bull market). Instead, it performs equally well at identifying both positive bullish indicators and negative bearish signals from the text.

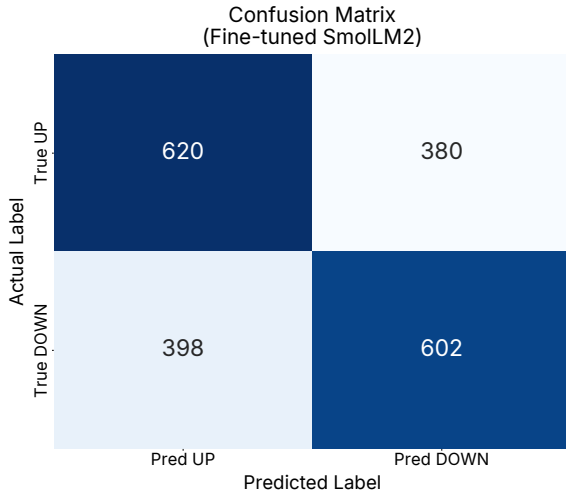


Fig. 3: Confusion matrix for the fine-tuned SmoLLM2-360M model. The high degree of symmetry across the main diagonal highlights the model’s balanced performance and lack of directional bias.

E. Rigorous Statistical Significance Testing

To ensure that our observed accuracy of 61.1% was not merely a statistical artifact of the specific test sample, we conducted a formal hypothesis test. Utilizing a two-tailed binomial exact test with a null hypothesis of $H_0 : p = 0.5$ (where p is the true probability of a correct prediction, equivalent to random guessing in a balanced dataset):

- **SmoLLM2-360M Base:** Yielded p -value > 0.05 . We fail to reject the null hypothesis; the base model is guessing randomly.
- **Qwen2.5-0.5B:** Yielded a marginally significant p -value ≈ 0.004 . The larger model performs slightly, but significantly, better than random.
- **SmoLLM2-360M Tuned:** Yielded a p -value < 0.00001 . We strongly reject the null hypothesis.

Given the robust sample size of $N = 2,000$, the performance of the fine-tuned model is highly statistically significant, confirming that the learned textual-financial relationships are robust and reliable.

F. Qualitative Case Study

To better understand the model’s decision-making process, we manually examined specific instances where the model succeeded and failed.

Case 1: Subtle Sentiment (True Positive). In one instance, a news article discussed a company’s leadership transition. While the text contained seemingly negative words (e.g., “resignation,” “departure”), the context indicated that the departing CEO had been widely criticized by shareholders, and the replacement was highly regarded. The base model predicted “down,” likely keying on the negative vocabulary. The tuned model correctly predicted “up,” demonstrating an ability to parse broader contextual sentiment beyond superficial keyword matching.

Case 2: Forward-Looking Guidance (True Negative). Another article detailed an earnings report where the company technically “beat” consensus estimates for the current quarter but provided significantly reduced forward-looking guidance for the remainder of the fiscal year. The unspecialized model predicted “up,” focusing on the immediate earnings beat. The fine-tuned model correctly predicted “down,” recognizing that forward guidance often dictates market reaction more strongly than past performance.

Case 3: Unforeseen Macro Events (False Positive). In a failure case, the model predicted “up” based on a highly positive product launch announcement. However, the broader market experienced a massive, sector-wide sell-off the following day due to unexpected Federal Reserve interest rate commentary. This highlights a fundamental limitation: text-only models cannot account for exogenous macroeconomic shocks that override company-specific news.

V. DISCUSSION

The findings of this research offer profound implications for the future architecture of AI-driven financial analysis systems.

A. Mechanisms of Fine-Tuning Success

The striking disparity between the base and fine-tuned models' performance necessitates an exploration of why PEFT is so effective in this specific domain. We hypothesize three primary mechanisms driving this success:

- 1) **Latent Feature Alignment:** The base SmoLLM2-360M model already possesses a strong foundational understanding of English syntax, grammar, and general semantics. Fine-tuning does not teach the model language; rather, it forcefully aligns these existing, latent linguistic representations with highly specific market-relevant outcomes.
- 2) **Complex Pattern Specialization:** Financial news is characterized by recurring, highly formalized narrative structures (e.g., earnings beats/misses, FDA trial approvals, C-suite restructuring, M&A rumors). The fine-tuning process allows the model to map these complex, multi-sentence narrative patterns directly to historical price reactions, acting effectively as an automated, high-dimensional pattern matching engine.
- 3) **Task-Specific Adaptation:** Standard language models are typically trained on causal language modeling (predicting the next token) to generate fluent text. The instruction-based fine-tuning format acts as a powerful constraint, forcing the model to abandon general text generation and focus entirely on synthesizing the provided context into a singular, binary discriminatory decision.

B. The Paradigm Shift: Specialization over Massive Scale

Historically, the NLP community has operated under the assumption of a "bitter lesson," suggesting that relentlessly scaling up parameter counts and compute is the primary path to superior performance. Our results challenge this notion in specialized commercial domains.

The fine-tuned SmoLLM2-360M unequivocally outperforms the larger Qwen2.5-0.5B, despite operating with approximately 28% fewer parameters. This provides compelling evidence that:

- Broad, encyclopedic knowledge encoded in massive general-purpose models (like knowing capital cities or how to write Python code) is largely irrelevant, and perhaps even distracting, when tasked with narrow domain-specific problems.
- Intensive specialization through LoRA and high-quality data can overcome significant architectural and parameter-count disadvantages.
- Financial prediction tasks rely heavily on specific vocabulary and situational logic, which are best acquired through targeted training rather than generalized pre-training.

C. Practical Deployment Implications for Algorithmic Trading

For quantitative hedge funds, high-frequency trading (HFT) firms, and retail algorithmic traders, the deployment of a

highly accurate, yet extremely small language model offers transformative operational advantages:

- **Ultra-Low Latency Inference:** In HFT, nanoseconds dictate profitability. A 360M parameter model, especially when quantized and deployed on specialized AI accelerators or FPGAs, can process news articles and generate predictions in a fraction of the time required by billion-parameter models, granting a crucial speed advantage in order execution.
- **Drastically Reduced Hardware Costs:** Running inferences on 70B parameter models requires arrays of highly expensive, high-power server GPUs (e.g., NVIDIA H100s). A 360M parameter model comfortably operates on commodity hardware, edge devices, or inexpensive cloud instances, radically reducing operating expenditures.
- **Distributed Edge Deployment:** The small memory footprint allows these models to be deployed directly on localized exchange co-location servers, minimizing network transmission latency.

D. Limitations and Constraints

While our results are highly promising, several critical limitations must be explicitly acknowledged before translating these findings into actionable trading strategies:

- 1) **Simplification to Binary Classification:** Real-world market movements are continuous, not discrete. By simplifying the problem to a binary "up/down" classification, the model fails to predict the magnitude of the movement. A predicted "up" movement of 0.5% does not warrant the same risk exposure as a predicted "up" movement of 15%.
- 2) **The 5% Threshold Dependency:** Our chosen volatility threshold of $\pm 5\%$ is somewhat arbitrary and biases the dataset towards highly volatile stocks (often tech or biotech). The model's performance on blue-chip stocks with lower historical volatility remains unproven.
- 3) **Assumption of Market Inefficiency:** Our evaluation operates under the theoretical assumption that one can execute a trade exactly at the market open price. In reality, the market may "gap up" or "gap down" at the open, immediately pricing in the overnight news, thus eliminating the profitable spread before an order can be filled. Slippage, bid-ask spreads, and broker commissions further erode theoretical returns.
- 4) **Temporal Concept Drift:** Financial markets are highly dynamic systems prone to regime changes (e.g., transitioning from a low-interest-rate tech boom to a high-inflation environment). A model trained on historical data may rapidly lose predictive validity as macroeconomic conditions and regulatory environments evolve, a phenomenon known as concept drift.
- 5) **Survivorship Bias:** Our dataset was constructed using currently active NASDAQ components. This methodology inherently introduces survivorship bias, as it excludes companies that generated overwhelmingly neg-

ative news and subsequently went bankrupt or were delisted during the historical period.

E. Future Research Trajectories

This study establishes a foundational proof-of-concept, but leaves substantial room for expansion. Future research should prioritize:

- **Regression-Based Magnitude Prediction:** Transitioning the architecture to predict continuous expected returns rather than discrete classes, allowing for more nuanced risk-reward calculations.
- **Multi-Modal and Temporal Fusion:** Integrating the textual news pipeline with simultaneous streams of numerical time-series data (e.g., technical indicators, moving averages, trading volume) to create a more holistic, multi-modal forecasting engine.
- **Mechanistic Interpretability:** Employing techniques like attention rollout and layer-wise relevance propagation to visualize precisely which sentences, phrases, or specific words in the news article triggered the model’s ultimate prediction. This transparency is crucial for regulatory compliance in finance.
- **Reinforcement Learning from Financial Feedback (RLFF):** Training models not just to predict direction, but to maximize an actual simulated portfolio objective function (like the Sharpe Ratio or Sortino Ratio) over time, punishing risky trades and rewarding consistent, risk-adjusted returns.

VI. CONCLUSION

This paper rigorously investigated the application of parameter-efficient small language models to the challenging task of predicting short-term equity price movements based on unstructured financial news. Our empirical findings conclusively demonstrate that the SmolLM2-360M architecture, when subjected to targeted LoRA fine-tuning, achieves a highly significant accuracy of 61.1% and a Matthews Correlation Coefficient of 0.220 on a difficult, thresholded dataset.

Crucially, this fine-tuned SLM dramatically outperformed not only its own base configuration but also a larger, more resource-intensive generalized language model. These results validate the hypothesis that in specialized, jargon-heavy domains like high-stakes finance, deep task-specific adaptation via high-quality data is vastly superior to reliance on the broad, generalized knowledge embedded in massive parameter counts.

The implications for the quantitative finance industry are substantial. By proving that robust predictive signals can be extracted using models with fewer than half a billion parameters, we highlight a pathway toward ultra-low latency, highly cost-effective, and edge-deployable AI trading systems. While significant engineering hurdles regarding slippage, market impact, and risk management remain before fully autonomous deployment is viable, this research establishes that small, specialized models represent a powerful, highly efficient

new tool in the algorithmic trader’s arsenal. As the open-source community continues to refine SLM architectures, their role in rapidly parsing and acting upon the world’s financial information stream will undoubtedly accelerate.

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